



BAHIR DAR UNIVERSITY
BAHIR DAR INSTITUTE OF TECHNOLOGY
FACULTY OF MECHANICAL AND INDUSTRIAL
ENGINEERING

Course Guidebook for Engineering Thermodynamics I

1. GENERAL INFORMATION OF THE COURSE

Course Name:	Engineering Thermodynamics I
Course Code:	Meng 2132
Course Credits Points:	5 hrs
Contact Hours:	2 hrs lecture, 2hrs Tutorial and 1hr lab per week
Course Module:	Engineering Mechanics
Status of Course:	Compulsory
Pre-Requisites:	None
Students' Department:	Mechanical Engineering
Degree Program:	BSc in Mechanical Engineering
Students Admission:	Regular
Year of Study:	2 nd
Semester:	Second Semester
Academic Year:	2024/2025
Course Duration:	From March 2025 to June 2025
Course Chair:	Dr. Bimrew Tamrat
Course Instructors:	Mr. Endedilu Shumet (MSc in Thermal Engineering) Email: endeix37@gmail.com Office: Tana: 203
Consultation:	Saturday 2:00 PM -4:30 PM

2. COURSE DESCRIPTION

Introductory concepts and definitions. Properties of pure substances. Energy transport by work, heat and mass. The first law of thermodynamics and energy. The second law of thermodynamics. Entropy. Availability and Irreversibility.

3. OBJECTIVES OF THE COURSE

At the end of the course, students will be able to:

- Understand the relationship between internal energy, heat and work as expressed by the First Law of Thermodynamics.
- Apply the conservation of energy to thermodynamic systems
- State and explain the Second Law of Thermodynamics
- Explain how the Carnot cycle applies to heat engines and refrigeration cycles
- Explain the concept of entropy.

- Apply the concept of availability, irreversibility and the second law in defining the efficiency of a system

4. COURSE LEARNING OUTCOMES (CLO)

At the end of the course, students will be able to:

No	Course Learning Outcome (CLO)	Performance Indicators (PIs)	Program Student Outcome (PSO)
1	Apply and Understand Basic Thermodynamic Principle and laws of thermodynamics to solve simple engineering problems.	PI 1.1 - 1.4,	1
2	Apply problem solving technique in the analysis of various engineering devices involving closed and open systems by Applying the First Law of Thermodynamics	PI 1.1 - 1.4, PI 2.1,2.2	1,2
3	Analyze engineering systems related to heat transfer, work, change in entropy and second law of thermodynamics	PI 1.1 - 1.4, PI 2.1,2.2 PI 7.1 7.2	1,2,7
4	Evaluate thermodynamics cycles efficiencies of simple thermal systems	PI 3.1 - 3.4 PI 6.1 - 6.4	1,6

5. THE COURSE TIME SCHEDULE AND TEACHING METHODOLOGY

Time Schedule	Course Contents	Teaching Methodology and	Reference	PSO
Week 1	Chapter 1: Introductory Concepts & Definitions: 1.1. Introduction, Thermodynamic systems; 1.2. Property, state, phase, process and equilibrium; 1.3. Dimensions and units; 1.4. Some basic properties of thermodynamics.	Lecture Tutorial Group discussion	Cengel Y A., Bole M A., Thermodynamics – An Engineering Approach, Sep 22, 2006. Page 1-17	1
Week 2-3	Chapter 2. Pure Substances: 2.1. Properties of pure substances; 2.2 Equilibrium mixture of vapor-liquid-phase, phase diagrams; 2.3. Independent properties of pure substances; 2.4 Thermodynamic property tables; 2.5. Equation of state.	Lecture Tutorial Group discussion Reading assignment	Cengel Y A.,Bole M A., Thermodynamics – An Engineering Approach, Sep 22, 2006. Page 111-144	1

Week 4-5	Chapter 3. Work and Heat: 3.1 Work; units of work; 3.2 Expansion and compression work; 3.3 Other forms of work; 3.4 Heat; units of heat.	Lecture Tutorial Group discussion Reading assignment QUIZ	Cengel Y A.Bole M A., Thermodynamics – An Engineering Approach, Sep 22, 2006. Page 165-193	1,2
Week 6-7	Chapter 4. The First Law of Thermodynamics and Energy: 4.1. The first law and the closed system; 4.2. The first law and the control volume.	Lecture Tutorial Group discussion Reading assignment	Cengel Y A.Bole M A., Thermodynamics – An Engineering Approach, Sep 22, 2006.	1,2
Week 8	Mid Exam			
Week - 9-10	Chapter 5. The Second Law of Thermodynamics: 5.1. Basic concepts; 5.2. Heat engines and refrigerators; 5.3. Statements of the second law of thermodynamics; 5.3. Reversible and irreversible processes; 5.4. The Carnot cycles; 5.5. Thermodynamic temperature scale.	Lecture Tutorial Group discussion Reading assignment	Cengel Y A.,Bole M A., Thermodynamics – An Engineering Approach, Sep 22, 2006. Page 51-78	1,2,6
Week 11-12	Chapter 6. Consequences of The Second Law of Thermodynamics and Entropy: 6.1. Entropy and the Clausius inequality; 6.2. Definition of entropy; 6.3. Definition of the second law of thermodynamics; 6.4 Heat transfer as the area under the T-s curve 6.5. Isothermal, Reversible process 6.6. Isentropic process and Isentropic efficiency.	Lecture Tutorial Group discussion Reading assignment	Cengel Y A.,Bole M A., Thermodynamics – An Engineering Approach, Sep 22, 2006. Page 279-315	1,2
Week 13-14	Chapter7.Availability and Irreversibility: 7.1. Introduction; 7.2. Control volume undergoing a steady state process; 7.3. Control volume undergoing unsteady state process; 7.4. Availability; second law efficiency.	Lecture Tutorial Group discussion Reading assignment Assignment	Cengel Y A.,Bole M A., Thermodynamics – An Engineering Approach, Sep 22, 2006. Page 331-378	1,2,7
15-16	Final Exam			

6. ASSESSMENT AND EVALUATION

- The course utilizes a variety of continuous assessment and evaluation methods to ensure the successful achievement of course learning outcomes. These methods include assignments, projects, quizzes, midterm exams, final exams, homework, classwork, presentations, and more. It is imperative that all assignments and tasks are completed within the specified deadlines. If you encounter any difficulties with the assignments or tests, please contact the instructor promptly for assistance.
- **Assignments, reading assignments, and project (if any):** Assignments may involve collaborative work for brainstorming and developing general approaches. However, unless specified otherwise, each individual must complete his or her own submissions. Assignments and projects may entail both individual and group components, with equal participation expected from all group members. While collaboration is encouraged, submitting another individual's work as one's own constitutes academic dishonesty and will incur penalties. All group members must contribute equally. Timely submission is mandatory, as late submissions will not be graded. Oral exams will accompany assignments and projects, requiring thorough preparation from all members. Additionally, the instructor reserves the right to assign additional tasks or assignments as necessary.
- **Quizzes:** Quizzes serve to assess students' current understanding of the coursework. They are conducted in class following lectures or tutorial sessions.
- **Mid-term Exam and Final Exam:** The exams will be closed book. During an exam, all necessary materials will be provided by the examiner.

Table 1. Summary of Assessments

No	Assessment type	Mark allotted	Content Coverage	Time Schedule
1	Quiz	10%	Chapter 2 & 3	Week 4
2	Mid exam	25%	Chapter 1, 2, 3 & 4	Week 8
3	LAB	5%	Refrigeration Cycle	Week9
4	Assignment	10%	Chapter 5, 6, and 7	Week 11
5	Final Exam	50%	Chapter 5, 6, & 7	Week 15
	Total	100%		

7. COURSE POLICY

- ✓ All students are expected to adhere to the code of conduct for students as per the Senate Legislation of Bahir Dar University, December 2020, during the duration of this course. Academic misconduct, which includes cheating, fabrication, and plagiarism, will not be tolerated and will be promptly reported to the relevant authorities for appropriate action.
- ✓ The teaching methodology for this course will be diverse, encompassing activities such as reading, lectures, tutorials, discussions, homework, classwork exercises, presentations, etc. Active participation from students is expected, including asking questions and raising issues whenever clarification is needed.
- ✓ Regular attendance and punctuality are expected from all students. Those who miss more than 15% of the semester's classes will not be eligible to sit for the final exam. Attendance will be recorded for each class, and students are responsible for all classroom announcements and changes. Any student who exceeds the 15% absence threshold will receive a grade of 'NG' (No Grade) and must provide acceptable reasons for their absence, as per Senate legislation, Art. 174 for NG management.
- ✓ Cell phones brought to class must be set to silent mode to avoid disruption to the teaching-learning process. Any form of disturbance during class time is unacceptable and must be avoided.
- ✓ All students are required to have complete records of course assessments and evaluations. If a student misses a quiz, midterm exam, or any other continuous assessment without a valid reason,

they will receive a score of zero. However, if a student fails to attend the final exam the instructor will record an "NG" according to senate legislation, Art. 171.

- ✓ It is important to carefully follow the instructions provided in your course guidebook to complete all course requirements.

8. GRADING SYSTEM

The grading scale of the course is fixed scale according to senate legislation of Art. 177.

Table 2. Grading scale of the course

Raw Mark Interval (100%)	Corresponding Letter Grade	Corresponding Fixed Number Grade
$90 \leq x \leq 100$	A ⁺	4.0
$85 \leq x < 90$	A	4.0
$80 \leq x < 85$	A ⁻	3.75
$75 \leq x < 80$	B ⁺	3.50
$70 \leq x < 75$	B	3.00
$65 \leq x < 70$	B ⁻	2.75
$60 \leq x < 65$	C ⁺	2.50
$50 \leq x < 60$	C	2.00
$45 \leq x < 50$	C ⁻	1.75
$40 \leq x < 45$	D	1.00
$x < 40$	F	0.00

9. Text and reference book

A. Text Book

1. Cengel Y A.,Bole M A., Thermodynamics - An Engineering Approach, Sep 22, 2006.

B. Reference Materials

1. Sonntag R.E.,” Fundamentals of Thermodynamics”, Sept 13, 2004.
2. Michael J. Moran, H.N. Shapiro, “Fundamentals of Engineering Thermodynamics”, Mar 9, 2007.
3. Eastop T.D and McConkey A., Applied Thermodynamics, Feb 29, 1996.
4. WarkK.Jr, Advanced Thermodynamics for Engineers, Sep. 1994

10. APPROVAL

The following bodies confirmed the course guidebook accordingly.

Position	Name	Signature	Date
Course Instructors	Endedilu Shumet		20/06/2017 E.C
Course Module Chair	Abraham Adera		20/06/2017 E.C
Chair Holder (Thermal Engineering)	Dr. Bimrew Tamrat		20/06/2017 E.C
V/Dean	Abebe Genetu		20/06/2017 E.C

